

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Winter Examination - 2022

Course: B. Tech.

Branch :

Semester : III

Subject Code & Name: Engineering Mathematics(BTBS301)

Duration: 3 Hr.

Max Marks: 60

Date: / /20

Instructions to the Students:

1. All the questions are compulsory
2. The level of question-expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Find the Laplace transform of $e^{-2t} \int_0^t \frac{\cos 2t}{t} dt$.	Understand	6
B) Find the Laplace transform of the periodic function, $f(t) = \frac{t}{T}$ for $0 < t < T$, & $f(t+T) = f(t)$.	Understand	6
C) By using Laplace transform, evaluate $\int_0^\infty e^{-2t} t^2 \sin 3t dt$	Evaluation	6
Q.2 Solve Any Two of the following.		12
A) By using convolution theorem, find inverse Laplace transform of $\frac{s}{(s^2+1)(s^2+4)}$	Application	6
B) Find inverse Laplace transform of $\cot^{-1}\left(\frac{s+3}{2}\right)$	Application	6
C) Using Laplace Transform, solve $\frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 3y = e^{-t}$, given $y(0) = 1$ & $y'(0) = 0$	Application	6
Q.3 Solve Any Two of the following.		12
Using the Fourier integral representation, show that		
A) i) $\int_0^\infty \frac{\omega \sin x \omega}{1+\omega^2} d\omega = \frac{\pi}{2} e^{-x} (x > 0)$ ii) $\int_0^\infty \frac{\cos x \omega}{1+\omega^2} d\omega = \frac{\pi}{2} e^{-x} (x \geq 0)$	Understand	6

Find the Fourier transform of the function m) $f(x) = \begin{cases} 1 - x^2 & x \leq 1 \\ 0 & x > 1 \end{cases}$ Hence evaluate $\int_0^{\infty} \frac{x \cos x - \sin x}{x^4} \cos \frac{x}{2} dx$	Evaluation	6
Using Parseval's identity, show that c) $\int_0^{\infty} \frac{t^2}{(4+t^2)(9+t^2)} dt = \frac{\pi}{10}$	Application	6
Q.4 Solve Any Two of the following. 12		
A) Form the partial differential equation by eliminating arbitrary function from $f(x + y + z, x^2 + y^2 + z^2) = 0$	Understand	6
B) Solve the partial differential equation $(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$	Application	6
C) Find the temperature in a bar of length 2 units whose ends are kept at zero temperature & lateral surface insulated if the initial temperature is $\sin \frac{\pi x}{2} + 3 \sin \frac{5\pi x}{2}$	Application	6
Q.5 Solve Any Two of the following. https://www.batuonline.com 12		
A) Prove that the function $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ is a harmonic function & hence determine the corresponding analytic function, $f(z) = u + iv$.	Understand	6
B) Evaluate, by using Cauchy's integral formula: i) $\oint_C \frac{e^{-z}}{z+1} dz$, where C is the circle $z = 2$. ii) $\oint_C \frac{\sin^2 z}{(z-\frac{\pi}{6})^3} dz$, where C is the circle $z = 1$.	Evaluation	6
C) Evaluate $\int_C \frac{2z-1}{z(z+1)(z-3)} dz$, where C is the circle $z = 2$	Evaluation	6
*** End ***		